

Yiming Jia

925-791-0921 | jasonjiaym@gmail.com | linkedin.com/in/yiming-jia-0a10a9258 | github.com/TrashyTuber | yimingjia.dev

EDUCATION

Northwestern University

Evanston, IL

B.A. in Computer Science & B.M. in Music Composition; GPA: 3.86

- **Relevant Coursework:** Fundamentals of Computer Programming II (C/C++), Data Structures & Algorithms, Machine Learning, Human Computer Interactions

EXPERIENCE

Trajecta

Remote

Co-Founder & Lead Engineer

Apr 2026 – Present

- Built and deployed an AI college-admissions planning platform with 100+ active users as an alternative to private consultants (React/Vite, Express/TypeScript on Render, Supabase Postgres & auth)
- Architected a multi-agent LLM admissions committee: four reader agents (academic, narrative, impact, context) issue schema-constrained verdicts in sequence, and a chair agent decides admit/waitlist/reject from their reports
- Built an eval harness on real applicant outcomes: 82.6% admit vs. non-admit accuracy across 201 profile-school pairs, against a 65% always-reject baseline
- Built a PDF import pipeline that treats uploads as untrusted: file validation before model access, schema-constrained extraction, and sanitized, capped output, with no document storage
- Added GitHub Actions CI (lint, type-check, 78 Vitest test files), Stripe subscription billing, and LLM rate limiting

TIDAL Lab, Northwestern University

Evanston, IL

Undergraduate Research Assistant (PI: Prof. Michael Horn)

Oct 2026 – Present

- Contributing to TunePad, an NSF-funded browser platform where Evanston 5th–8th graders learn to code by composing music in Python
- Implementing real-time envelope and reverb effects, and porting the DAW-style track view to the classroom version

Develop & Innovate for Social Change (DISC)

Evanston, IL

Front-end Engineer

Jan 2026 – Jun 2026

- Built the React front end of a client admin platform replacing Google Forms, Sheets, Zapier, and Looker Studio, managing intake data for 400+ participants across two programs, including Oakton Community College
- Developed an editable database view where staff update participant records (enrollment, attendance, certification, employment, wages), replacing the client's shared spreadsheets
- Built a Recharts outcomes dashboard on live data and a templated participant email system; team of six

PROJECTS

Melody Harmonizer | *Python, PyTorch, music21, Gradio* | [Live Demo](#)

Aug 2026 – Oct 2026

- Built a BiLSTM sequence-labeling model that predicts a chord for every melody note (Nottingham folk dataset), using a transposition-closed 121-chord vocabulary and 12-key augmentation split by song to prevent leakage
- Benchmarked it against a size-matched Transformer encoder over 3 seeds; the BiLSTM led on every metric and reached 66.8% exact and 76.8% key-aware functional accuracy on a held-out test set vs. a 25% baseline
- Added a learned chord-transition Viterbi decoder that brought the chord-change rate within 5% of ground truth on test songs (from 8% over); deployed as a Gradio app on Hugging Face Spaces with MIDI and audio output

Prophet Hacks: Prediction-Market Trading Bot | *Python*

May 2026

- Built a trading system with a two-stage LLM architecture (scout proposes probabilities, judge vetoes overconfident trades), fractional-Kelly sizing, and per-market exposure limits
- Routed markets to external data (FRED, Open-Meteo, news, sports odds) and wrote a backtesting harness over 1,200 markets reporting per-category Brier scores

Algorithmic Live Coding Performance | *JavaScript, Strudel*

Nov 2025 – Present

- Won 2nd place at the BitCrush Hackathon with a live-coded musical performance generating patterns in real time
- Recording passed 250,000 Instagram views; now producing a 4-track EP in code with an entertainment company

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, C#, SQL, HTML/CSS

Frameworks & Tools: React, Node.js, Express, Supabase/Postgres, Docker, GitHub Actions, Vitest, Render

Data & ML: PyTorch, scikit-learn, Pandas, NumPy, LSTMs, Transformers, Gradio, Hugging Face, Weights & Biases